# Measurement Map of the Theory Paper

Companion to `theory/wealth_garch_theory.tex`, 23 October 2026

## 1 Scope

The map names every object of the model of the theory paper once, and the manuscript uses these names and no others. The names are those of the Names table at the end of this document, agreed with the author on 10, 13 and 19 October 2026, and four names the author accepted on 23 October 2026, the solvency floor, the run-off rate, the liquidity haircut and the value function. Inputs are what the model takes as given, internal objects are built from the inputs inside the model, and outputs are what the rows predict. The column Directly observed says Yes when a record of the world gives the object without any computation, which is when its last column begins with Recorded, and No otherwise. The other status words are Elicited, Computed, Assumed, Never observed and A device. Every symbol in the manuscript's table of assumptions has exactly one row here. Where a row names the panel, it is the FDIC panel of US banks, 2015 to 2024, that the author's earlier empirical work read, and the figures are from that reading. Lean names are theorems of `theory/lean/`, audited by `verify_theory.py`.

The theory paper is a theory of what a bank's policy would reveal if the bank followed a threshold policy, and the map says what that leaves for measurement. The asymmetry parameter and the fixed issuance cost are never observed. The two thresholds that identify them are recorded as the state at a dividend and at an issue, and on the panel read neither behaves as a threshold. The limits of buffer variance are computed from assumed volatilities and apply where the cap binds, and on that panel it does not. What reaches the data is the bound of M12 and the two recorded thresholds as events.

### 1.1 Inputs

Table 1: Inputs of the model.

| Symbol | Name | Type | Represents | Directly observed | Obtained how |
|---|---|---|---|---|---|
| $x$ | **Assets-to-liabilities ratio** | A real number above one. | Total assets over total liabilities, the state of the model (A1). | Yes | Recorded as total assets over total liabilities in the Call Report, `ASSET` over `LIAB`. |
| $\pi$ | **Risky exposure** | A function of the state with values between zero and the cap over the state. | The share of assets in the risky asset (A1, A2). No row derives it; M11 takes it at the cap and M12 bounds every compliant choice. | Yes | Recorded as risk-weighted assets over total assets, `RWAJT` over `ASSET`, under risk weights of zero on the riskless and one on the risky asset. On the panel its median is 0.69 against a cap of 2.38 at the median state, so the cap does not bind there. |
| $D$ | **Dividends** | A non-decreasing process, zero before the first date. | Cumulative payments to shareholders per unit of liabilities, the singular control of the bank's problem. Under the optimal policy it rises only when the state is at the dividend barrier (A6). | Yes | Recorded as cash dividends declared, `EQCDIV`, over liabilities. |

| Symbol | Name | Type | Represents | Directly observed | Obtained how |
|---|---|---|---|---|---|
| $I_n$ | **Recapitalisation** | A positive real number for each issue. | Capital raised in the $n$-th issue per unit of liabilities, the impulse control, at the issue time $\tau_n$. Under the optimal policy it is the injection target less the recapitalisation trigger (A6). | Yes | Recorded as net stock issuance (`EQCSXQ`) and capital from the parent holding company (`EQCBHCTR`), over liabilities. |
| $r$ | **Riskless rate** | A positive real number. | The return on the riskless asset, in the drift of the state (A1). | No | Assumed at 0.025 in the paper's illustrations and at 0.02, the baseline of **?**, in its comparative statics. |
| $\mu_s$ | **Expected asset return** | A real number. | The drift of the risky asset, in the drift of the state (A1). | No | Assumed at the baseline of **?**, 0.04. |
| $\mu_L$ | **Liability drift** | A real number. | The drift of liabilities, in the drift of the state and in the liability-net discount rate, which is the discount rate less the liability drift (A1, A4). | No | Assumed at the baseline of **?**, 0.03. |
| $\gamma$ | **Net income on liabilities** | A real number. | Income earned on liabilities net of their cost, per unit of liabilities, in the drift of the state (A1). | No | Assumed at the baseline of **?**, 0.02. They read the same term as the exogenous growth of deposits, which adds to assets. |
| $\sigma$ | **Asset volatility** | A positive real number. | The volatility of the risky asset, and the letter of the diffusion of the state (A1, A3). | No | Assumed at the baseline of **?**, 0.08. The volatility of the book ratio is computed on the panel and is 4.1 times smaller than the model's at the recorded exposure, so it is not this object. |
| $\sigma_L$ | **Liability volatility** | A positive real number. | The volatility of the shock to liabilities (A1). | No | Assumed at the baseline of **?**, 0.03. |
| $c$ | **Return-liability correlation** | A real number of absolute value below one. | The correlation between the risky return and the liability shock (A1). | No | Assumed at the baseline of **?**, 0.20. |
| $q$ | **Required capital** | A positive real number. | The capital regulation requires, as a ratio to liabilities, the zero of the capital buffer (A1). | No | Assumed. A floor on the leverage ratio gives a candidate, 3/97 for the 3% Tier 1 leverage ratio when Tier 1 equals book equity, and the paper does not fix it. Buffer variance does not depend on it. |
| $a_1$ | **Solvency floor** | A real number between zero and the liquidity haircut. | The floor on equity over risky assets, the slope of the solvency branch of the regulatory cap (A2). | Yes | Recorded in the Basel standard as the CET1 minimum of 4.5% of risk-weighted assets, the baseline of **?**; which tier it stands for is not fixed. |
| $a_2$ | **Run-off rate** | A real number below one. | Net 30-day outflows as a share of liabilities, the intercept of the liquidity branch (A2). | No | Assumed at 0.05, the standard's run-off factor for stable retail deposits. The standard sets a factor per class of liability, so one rate for all liabilities is the model's reduction. |
| $a_3$ | **Liquidity haircut** | A real number between the solvency floor and one. | The haircut on the risky asset in high-quality liquid assets, the slope of the liquidity branch (A2). | No | Assumed at 0.30, the baseline of **?**, which is none of the standard's haircuts. |
| $\rho_L$ | **Liability-net discount rate** | A positive real number. | The discount rate of the problem written per unit of liabilities (A4). | No | Assumed from the shareholders' discount rate, 0.12 at the baseline of **?**. |

| Symbol | Name | Type | Represents | Directly observed | Obtained how |
|---|---|---|---|---|---|
| $\lambda_S$ | **Asymmetry parameter** | A real number at least one. | How much more the objective weighs a shortfall below the reference than a surplus above it (A4). At one the shortfall penalty vanishes and the model has issuance costs alone (M8). | No | Never observed. It enters the rows through the shortfall penalty only. M3 identifies it locally with the fixed issuance cost from the dividend barrier and the recapitalisation trigger, a statement within the model, since on the panel neither threshold is measured as one (the rows of the dividend barrier and the recapitalisation trigger below). |
| $K$ | **Fixed issuance cost** | A positive real number. | The cost of an issue that does not scale with its size, per unit of liabilities (A5). | No | Never observed. Its counterpart is the part of an issue's cost that does not scale with the amount raised, which **?** set from implied issue cost and frequency rather than record (L8). M3 identifies it locally with the asymmetry parameter from the two thresholds. |
| $\kappa$ | **Proportional issuance cost** | A non-negative real number. | The cost of an issue per unit raised (A5). | No | Assumed at the baseline of **?**, 0.01 for dilution and 0.02 per unit issued. |

## 1.2 Internal objects

Table 2: Internal objects of the model.

| Symbol | Name | Type | Represents | Directly observed | Obtained how |
|---|---|---|---|---|---|
| $z$ | **Capital buffer** | A real number, $z = \log((x-1)/q)$. | The log of equity over required capital, zero where equity equals required capital and falling without limit towards the distress boundary (A1, M9). | No | Computed from the assets-to-liabilities ratio and required capital by its formula, whose inverse is `CapGeometry.zmap_inverse` and which is zero where equity equals required capital, `Coordinate.zmap_at_requirement`. The author's earlier empirical work records its discrete-time form as Tier 1 capital over 8.5% of risk-weighted assets. |

| Symbol | Name | Type | Represents | Directly observed | Obtained how |
|---|---|---|---|---|---|
| $R$ | **Reference** | A real number, $R = 1 + q$. | The state at which equity equals required capital, below which the shortfall penalty applies (A4). The capital buffer is zero there. | No | Computed from required capital by its formula. The capital buffer vanishes at it, `Coordinate.zmap_at_requirement`; the penalty is zero at or above it, `QVI_Part1.shortfall_zero_at_or_above_reference`, and positive below it for an asymmetry parameter above one, `QVI_Part1.shortfall_pos_below_reference`. Its counterpart is the required capital of the solvency floor's row, the level at which prompt corrective action's restrictions begin. |
| $u$ | **Regulatory cap** | A function of the state, $u(x) = \min\{(x-1)/a_1, (x-a_2)/a_3\}$. | The largest risky exposure regulation allows at each state, the smaller of its two branches (A2). | No | Computed from the assets-to-liabilities ratio and the cap geometry by its formula. The solvency branch is the smaller up to the crossing point, `CapGeometry.solvency_binds`, and the liquidity branch beyond it, `CapGeometry.liquidity_binds`. |
| $\bar{x}$ | **Crossing point** | A real number above one, $\bar{x} - 1 = a_1(1-a_2)/(a_3 - a_1)$. | The state at which the two branches of the regulatory cap are equal (M10). | No | Computed from the cap geometry by its formula, `CapGeometry.xbar_sub_one`. |
| $\mu$ | **Drift of the state** | A function of the state and the risky exposure, $\mu(x,\pi) = x((1-\pi)r + \pi\mu_s - \mu_L) + \gamma$. | The expected change per unit of time of the assets-to-liabilities ratio between issues and dividends (A1). Along the cap it is the drift whose derivative at the dividend barrier enters M1. | No | Computed from the riskless rate, the expected asset return, the liability drift and the net income on liabilities. |
| $\Lambda$ | **Shortfall penalty** | A function of the state, $\Lambda(x) = (\lambda_S - 1)(R-x)^+$. | The term of the objective that charges a shortfall below the reference (A4). Zero when the asymmetry parameter is one, `QVI_Part1.lambda_asymmetry_vanishes_at_one`. | No | Never observed. It is a term of the objective, and whose cost it stands for is a reading, the shortfall-penalty paragraph of the manuscript's primitives appendix. |
| $V$ | **Value function** | A function of the state, twice continuously differentiable on the inaction region (A6). | The value of the bank under its best policy, per unit of liabilities. | No | Never observed. Its counterpart would be the market value of equity per unit of liabilities, which the panel does not carry; **?** fit their parameters to such values (L21). |
| $\mathscr{M}V$ | **Impulse operator** | A function of the state, the supremum over targets of the value after an issue net of both issuance costs. | The value of issuing at once to the best target (A6, M5, M7). | No | A device. It exists only inside the quasi-variational inequality and the proofs of M5 and M7. |
| $W$ | **Expected discounted shortfall** | A function of the state, $W = -\partial_{\lambda_s} V$. | The expected discounted shortfall below the reference under the optimal policy (A6). | No | A device. It enters M1 to M3 through the envelope identity only. |

| Symbol | Name | Type | Represents | Directly observed | Obtained how |
|---|---|---|---|---|---|
| $N$ | **Discounted injection count** | A function of the state, $N = -\partial_K V$. | The expected discounted number of issues under the optimal policy (A6). | No | A device. It enters M1 to M3 through the envelope identity only; its sample counterpart, a discounted count of recorded issues, is used by no row. |

## 1.3 Outputs

Table 3: Outputs of the model.

| Symbol | Name | Type | Represents | Directly observed | Obtained how |
|---|---|---|---|---|---|
| $y^*$ | **Dividend barrier** | A real number above one. | The state at which dividends are paid (A6). M1 gives its response to the asymmetry parameter and the fixed issuance cost. | Yes | Recorded as the state at which a dividend is paid, from dividend quarters (`EQCDIV`) and the recorded state. On the panel, 56% of bank-quarters below the solved barrier pay a dividend and payment falls as the state rises, so no barrier is measured there, and under the conservation buffer the recorded payouts reflect a rule as well as the barrier. |
| $x_L$ | **Recapitalisation trigger** | A real number above one and below the dividend barrier. | The state at which capital is raised (A6). M2 gives its response to the two parameters. | Yes | Recorded as the state just before a recapitalisation, from net stock issuance (`EQCSXQ`) and capital from the parent holding company (`EQCBHCTR`). On the panel, issues start at a mean state of 1.11 against a solved trigger of 1.03. |
| $y_{\text{post}}$ | **Injection target** | A real number between the trigger and the barrier. | The state an issue reaches (A6). M7 says it does not depend on where the issue starts. | Yes | Recorded as the state just after a recapitalisation. On the panel the post-issue state rises with the pre-issue state, against the single target of M7. |
| $\zeta$ | **Buffer variance** | A function of the state, $z'(x)^2$ times the diffusion $\sigma^2(x, \pi(x))$. | The variance per unit of time of the capital buffer (A3). | No | Computed from the capital buffer, the asset volatility, the liability volatility, the return-liability correlation and the risky exposure by its formula, `CapGeometry.zvol2_eq`. Not computed, since a level needs the asset volatility, which is assumed. |
| $\nu_1^2$ | **Deficit limit** | A real number, $\nu_1^2 = \sigma^2/a_1^2 - 2c\sigma\sigma_L/a_1 + \sigma_L^2$. | The limit of buffer variance at the distress boundary where the exposure binds the solvency branch (M10, M11). | No | Computed from the asset volatility, the liability volatility, the return-liability correlation and the solvency floor by its formula, `CapGeometry.deficit_limit`. It applies where the cap binds, which on the panel it does not (the row of the risky exposure). |

| Symbol | Name | Type | Represents | Directly observed | Obtained how |
|---|---|---|---|---|---|
| $v_3^2$ | **Surplus limit** | A real number, $v_3^2 = \sigma^2/a_3^2 - 2c\sigma\sigma_L/a_3 + \sigma_L^2$. | The limit of buffer variance deep in surplus where the exposure binds the liquidity branch (M11), which needs borrowing (C3). | No | Computed from the asset volatility, the liability volatility, the return-liability correlation and the liquidity haircut by its formula, `CapGeometry.surplus_limit`. It applies in the leveraged case only (C3). |

# 2 Names

Every name the manuscript may use for an object of the model, with its symbol, its meaning, its status and the names it replaces. Agreed names were agreed with the author on the dates given in the Scope. In use names are in use and not yet put to the author. A name in the last column is retired and appears in the manuscript only inside a literature row or a sentence that cites the paper using it (MM-11 and MM-12 of `checks/verify_measurement_map.py`). The ledger of the paper (`checks/ledger_theory.py`) names its nodes from this table.

Table 4: Names of the map.

| Name | Symbol | Meaning | Status | Replaces |
|---|---|---|---|---|
| bank | none | The institution that files the report, the agent whose policy the model solves. | Agreed | none |
| investors | none | Those who buy the bank's stock or receive its dividends, its parent holding company among them. | Agreed | none |
| insured creditors | none | Those whose claims on the bank deposit insurance covers, so that they lose nothing if it fails. | Agreed | none |
| uninsured creditors | none | Those whose claims on the bank no insurance covers, the lenders of wholesale funds and the holders of deposits above the insurance limit. | Agreed | none |
| supervisor | none | The bank's primary federal regulator, which supervises it and receives its reports. | Agreed | none |
| insurer | none | The deposit insurer, which insures the insured creditors' deposits. | Agreed | none |
| assets-to-liabilities ratio | $x$ | Total assets over total liabilities, the state of the model. | Agreed | none |
| equity | $X - L$ | Assets less liabilities. | In use | none |
| required capital | $Q$, $q$ | The capital regulation requires, in dollars or as a ratio to liabilities. | Agreed | none |
| capital buffer | $z$, $z_t$ | The log of equity over required capital. Zero when equity equals required capital, falling without limit towards the distress boundary. | Agreed | log coordinate |
| distress boundary | $x = 1$ | The state at which equity is zero. | In use | deep in deficit |
| deficit | $z < 0$ | Equity below required capital. | In use | none |
| surplus | $z > 0$ | Equity above required capital. | In use | none |
| risky exposure | $\pi$ | The share of assets held in the risky asset. | Agreed | none |
| regulatory cap | $u(x)$ | The largest risky exposure regulation allows at each state, the smaller of its two branches. | In use | none |
| solvency branch | $(x-1)/a_1$ | The branch of the regulatory cap set by capital. It binds near the distress boundary. | In use | none |

| Name | Symbol | Meaning | Status | Replaces |
|---|---|---|---|---|
| liquidity branch | $(x-a_2)/a_3$ | The branch of the regulatory cap set by liquidity. It binds deep in surplus. | In use | none |
| crossing point | $\bar{x}$ | The state at which the two branches are equal. | In use | none |
| cap geometry | $a_1, a_2, a_3$ | The three parameters of the regulatory cap, taken together. Never a preference. | Agreed | none |
| solvency floor | $a_1$ | The floor on equity over risky assets, the slope of the solvency branch. | Agreed | none |
| run-off rate | $a_2$ | Net 30-day outflows as a share of liabilities, the intercept of the liquidity branch. | Agreed | none |
| liquidity haircut | $a_3$ | The haircut on the risky asset in high-quality liquid assets, the slope of the liquidity branch. | Agreed | none |
| value function | $V$ | The value of the bank under its best policy, per unit of liabilities. | Agreed | none |
| asset volatility | $\sigma$ | The volatility of the risky asset. | Agreed | none |
| expected asset return | $\mu_s$ | The drift of the risky asset. | Agreed | none |
| riskless rate | $r$ | The return on the riskless asset. | In use | none |
| liability volatility | $\sigma_L$ | The volatility of the shock to liabilities. | Agreed | none |
| return-liability correlation | $c$ | The correlation between the risky return and the liability shock. | Agreed | none |
| net income on liabilities | $\gamma$ | Income earned on liabilities, net of their cost. | In use | none |
| drift of the state | $\mu(x,\pi)$ | The expected change per unit of time of the assets-to-liabilities ratio between issues and dividends. | In use | none |
| liability drift | $\mu_L$ | The drift of liabilities, $\gamma + r_L$. | In use | none |
| deposit rate | $r_L$ | The rate paid on liabilities. | In use | none |
| discount rate | $\rho$ | The shareholders' discount rate. | Agreed | none |
| liability-net discount rate | $\rho_L$ | The discount rate of the problem written per unit of liabilities. | Agreed | none |
| return on equity | $r_E$ | Net income over equity. | Agreed | none |
| dividends | $D$ | Payments to shareholders, the singular control. | Agreed | none |
| dividend barrier | $y^*$ | The state at which dividends are paid. | Agreed | payout barrier |
| recapitalisation | $I$ | Capital raised in one issue, the impulse control. | Agreed | none |
| recapitalisation trigger | $x_L$ | The state at which capital is raised. | Agreed | injection trigger, recapitalisation boundary |
| injection target | $y_{\text{post}}$ | The state an issue reaches. | Agreed | none |
| thresholds | none | The dividend barrier and the recapitalisation trigger, taken together. | Agreed | none |
| fixed issuance cost | $K$ | The cost of an issue that does not scale with its size, per unit of liabilities. | Agreed | none |
| proportional issuance cost | $\kappa$ | The cost of an issue per unit raised. | Agreed | none |
| asymmetry parameter | $\lambda_S$ | How much more the objective weighs a shortfall below the reference than a surplus above it. | Agreed | loss aversion |
| shortfall penalty | $\Lambda$ | The term of the objective that charges a shortfall below the reference. Zero when the asymmetry parameter is one. | In use | none |
| reference | $R$ | The level below which the shortfall penalty applies. | In use | none |
| expected discounted shortfall | $W$ | The expected discounted shortfall below the reference under the optimal policy. | In use | none |

| Name | Symbol | Meaning | Status | Replaces |
|---|---|---|---|---|
| discounted injection count | $N$ | The expected discounted number of issues under the optimal policy. | In use | none |
| impulse operator | $\mathscr{M}V$ | The value of issuing at once to the best target, net of both issuance costs. | In use | none |
| buffer variance | $\zeta$ | The variance per unit of time of the capital buffer. | Agreed | z-volatility, capital buffer's volatility |
| deficit limit | $\nu_1^2$ | The limit of buffer variance at the distress boundary. | Agreed | saturation limits |
| surplus limit | $\nu_3^2$ | The limit of buffer variance deep in surplus. | Agreed | saturation limits |
| variance ratio | $\lambda_V^4$, $\nu_1^2/\nu_3^2$ | The deficit limit over the surplus limit. | Agreed | volatility ratio |
| induced variance asymmetry | $\lambda_V$ | The asymmetry read off the variance ratio, distinct from the asymmetry parameter. | Agreed | none |
| saturation scale | $\theta$ | How sharply the variance of the discrete-time recursion moves between its deficit and surplus levels. | In use | none |
| growth rate of required capital | $g$ | The growth rate of required capital. | In use | none |
| retention ratio | $b$ | The share of net income retained. | In use | none |
| surplus payout rate | $\delta_p$ | The share of surplus paid out in a period. | In use | none |
| deficit persistence | $\phi^-$ | How much of a deficit remains after one period. | In use | none |
| surplus persistence | $\phi^+$ | How much of a surplus remains after one period. | In use | none |